

Honest reflections

*what 14,973 student–AI conversations taught us
(and what we built next)*

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Theme: Leveraging GenAI as a learning and support resource

A FRIDAY AFTERNOON, ONE OF OUR LABS

Risk assessment due Monday. Forty students.

Our simulation won't write it for them — so they open another tab.

STUDENT

Can you write me a risk assessment for a small retail business adopting AI?

AI

Sure. Here is a risk assessment covering data privacy, customer trust, supplier integration, staff training, and rollout cost... (response continues for 8 paragraphs)

STUDENT

Thanks!

Total elapsed time: 90 seconds. The assignment is now done.

And nothing was learned.

THE NUMBER IN OUR TITLE

What we first reported from 14,973 turns

Five semesters of unmediated student–AI conversation inside a self-hosted teaching simulation. No coaching, no observers — just what students did when no-one was watching.

1.4 / 5

average critical-thinking score

LLM-judged sample of 300 sessions

1 in 24

turns contained any pushback

even when the AI was demonstrably wrong

The dominant pattern looked like polite acceptance.

That finding motivated a grant, an ethics application, and a randomised trial. Then we looked again.

HONEST REFLECTION № 1

We looked again — and the story changed

The dataset grew to 19,504 turns across four model generations, and we rebuilt the analysis with more robust detectors. The pattern is not flat — engagement is rising.

EARLY DEPLOYMENTS

1.07 / 5

critical thinking (LLM-judged)

61%

of turns were pure delegation

CURRENT DEPLOYMENT

2.50 / 5

critical thinking (LLM-judged)

31%

of turns were pure delegation

Students are engaging more than our first analysis suggested — and more each year. An honest reflection means saying so.

What observation can't tell us

THE CONFOUND

Model capability, cohort, time, and growing AI familiarity all moved together.

Observational data cannot separate them. We can describe the rise; we cannot say what caused it — or how to produce more of it.

THE REMAINDER

Even at the recent peak, engagement is only moderate, highly variable — and still delegation-heavy for a substantial minority of students.

The average improved. The distribution did not close.

*So the observational study is no longer our motivation. It is our baseline —
the distribution a deliberate intervention has to beat.*

THE PREMISE

**AI literacy isn't about the tool.
It's about how you converse with it.**



Conversation, not delegation: the quality of AI-assisted work depends on whether you are directing the AI or accepting its output.

WHAT WE BUILT NEXT

Interfaces already nudge — we invert the intent

Designing behaviour into an interface is a twenty-year-old discipline (Fogg, Eyal, Zuboff). Frontier chatbots close almost every reply with an engagement hook: “Want me to draft the next section?”

ENGAGEMENT-MAXIMISING

Designed to extend the session

Trigger: end of AI response

Cue: “Want me to keep going?”

Reward: another fluent output

Outcome: time on platform

CRITICAL-ENGAGEMENT

Designed to slow acceptance

Trigger: end of AI response

Cue: “What's the weakest part of that?”

Reward: a sharper second pass

Outcome: capability in the user

Same design grammar. Opposite intent.

THE INTERVENTION

One line, appended below every AI response

Ten variants in a pool, drawn at random each turn to resist habituation. Appended server-side — the model never sees the nudge text.

“Does that match what you expected? If something seems off, tell me what seems wrong.”

“What's the weakest part of that response?”

“Does that account for the details in your task, or is it a generic answer?”

Three of ten. Every variant: invites pushback rather than confirmation, is domain-agnostic, and assumes no prior AI experience. Final pool wording signed off with co-investigators before collection.

THE DESIGN DECISION PEOPLE ASK ABOUT

First, we silence the model's own nudges

Frontier models are trained to keep users engaged. If their built-in follow-up hooks reached the control arm, we would be measuring two nudges at once.

SYSTEM PROMPT (BOTH ARMS)

“Answer the user's question directly. Do not offer to continue, elaborate, summarise, or suggest follow-up questions. Do not include engagement hooks of any kind.”

*The only structured cue in the trial is ours — and only the treatment arm sees it.
The control is silent.*

DEMO · LIVE IF THE ROOM AND WI-FI COOPERATE

chat.locolabo.org

a 60–90 second walkthrough

What to look for:

- The AI's response — clean, no follow-up suggestions, no engagement hooks
- The nudge appearing under the response — one line, randomly drawn from ten
- What happens when a student takes the bait and pushes back

The trial is live in Semester 2 labs right now — this is the production system, not a mock-up.

THE TRIAL

A randomised controlled trial, four disciplines

DESIGN	Individual-level RCT, randomised within unit — nudge vs silent control
UNITS	ISYS6018 · MGMT6076 · MKTG1000 · MGMT1003
DISCIPLINES	Information Systems · Supply Chain · Marketing · People Culture Organisations (PCO)
SAMPLE	~80–120 students (≈40–60 per arm), structured lab sessions
MODEL	Claude Sonnet 4.6 in both arms — model quality held constant
GOVERNANCE	HREC 83897 (low risk) · pre-registered on OSF before unblinding
FUNDING	Curtin NBF Learning & Teaching Grant 2026 · AUD \$20,000

Four disciplines so we can test whether the effect generalises beyond information systems. Randomising within unit compares two students who sat the same brief with the same lecturer.

WHAT WE MEASURE

Behaviour per turn, depth per thread, outcome per task

ENGAGEMENT RATE

Every turn coded on three binary dimensions: Follow-Up, Challenge, Extension. All three absent = delegation.

Primary outcome: challenge-weighted score ($FU + 2 \cdot CH + EX$), fixed before unblinding.

DIALOGUE DEPTH

New since the design was first shared: per-thread depth on a 1–5 cognitive-presence ladder, against thread sheets fixed per unit before coding.

Catches the failure mode a rate-only study would miss: more turns without deeper thinking.

TASK OUTCOME + EQUITY

Task scored 1–5 on a rubric, within unit only — a marketing brief and a supply-chain audit are not commensurable.

Equity indicators (low SES, first-in-family): voluntary, self-reported at consent, destroyed after analysis.

Mixed-effects model: condition fixed, unit random intercept, equity terms as interactions. Where rate and depth diverge, depth governs interpretation.

THE QUESTION WE CARE MOST ABOUT

Does it lift everyone — or close a gap?

AI interaction habits are unequally distributed on arrival. Students with fewer informal support networks get fewer chances to build productive habits before assessment.

UNIFORM UPLIFT

Same benefit for everyone

The nudge helps all students equally. The average rises. The gap is unchanged.

GAP-CLOSING UPLIFT

More benefit where the need is greater

The nudge helps under-scaffolded students most. The average rises. The gap closes.

Honest caveat: with ~80–120 students this analysis is likely underpowered, so it is registered as exploratory — a hypothesis for the next study, not a verdict from this one.

What has already gone wrong (and what it cost)

The pilot slipped

Planned May–June; ran in July. Every downstream date compressed, and the power analysis is later than we would like.

The window is one-shot

The four units run once in Semester 2. A flaw surfaced too late, or weak lab attendance, costs a full year — recruitment is the lever for statistical power.

Our first analysis over-claimed

The 14,973-turn 'polite acceptance' story softened with more data and better methods. That correction is in this talk on purpose.

Rate isn't understanding

A nudge could raise the number of follow-ups without deepening thought — which is why depth, not rate, governs interpretation.

WHERE WE ARE RIGHT NOW

A trial in progress — not a retrospective

Foundation

Grant · HREC 83897 · co-Is

Mar–Apr 2026

Pilot + instruments

Nudge pool · thread sheets ·
power

May–Jul 2026

Data collection

Sem 2 labs, four units

Jul–Oct 2026

 **WE ARE HERE**

Coding + analysis

FU/CH/EX · depth · pre-
registered

Nov–Dec 2026

Results + resources

Papers · educator pack

2027

THIS MONTH

Students are in labs with the live system as we speak. A research-in-progress paper (design + pre-registration) goes to ACIS 2026 in mid-August; results land early 2027.

TO LEAVE YOU WITH

**Every AI interface is already nudging its
users.**

The only question is toward what.



For discussion: what is the equivalent of this one-line nudge in the AI tools your students — and your staff — use every day? Who designs it, and who reviews it?

THANK YOU

The keep-asking team

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CONTACT & LINKS

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chat.locolabo.org

the live trial platform

github.com/michael-borck/keep-asking

open-source app, instruments, and nudge pool

PROJECT TEAM

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Scaffolding AI Interaction · Curtin NBF Learning & Teaching Grant 2026 (AUD \$20,000) · HREC 83897